Deduce the World: How Far Can Reasoning Go?

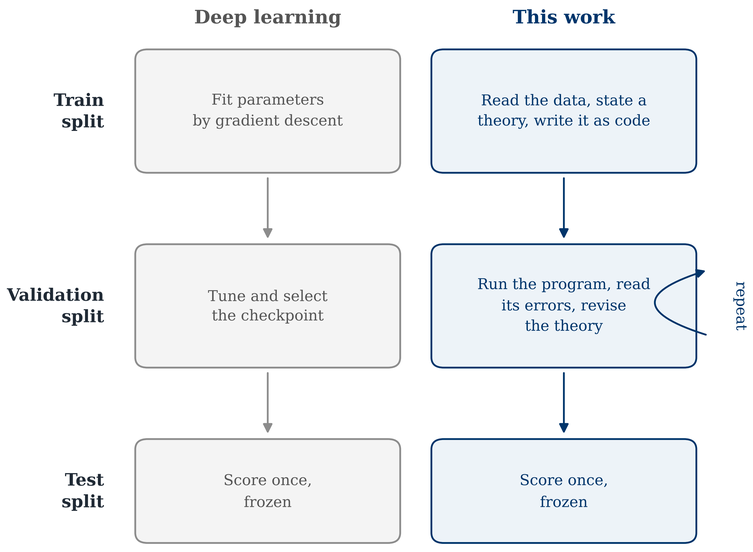
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Results & figures

1 Theories, not fitted predictors

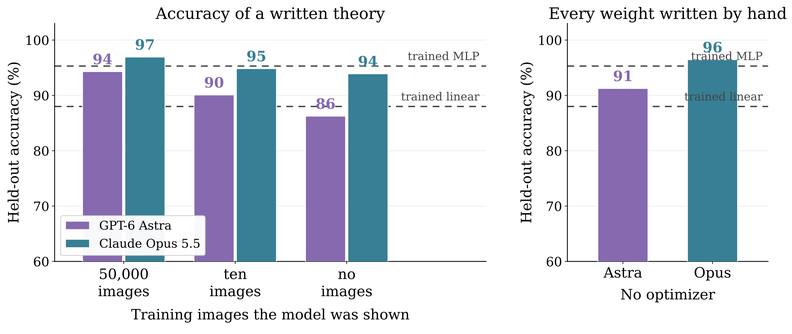
Science deduces the world from first principles, revising its theories where their predictions fail.

We ask whether a reasoning model can form scientific theories, and whether those theories generalize better than deep learning.



May	read papers and the web, measure the training data, inspect its own errors
May not	fit parameters to labels, search automatically over rules, store examples, see the test split

2 How much data a theory needs

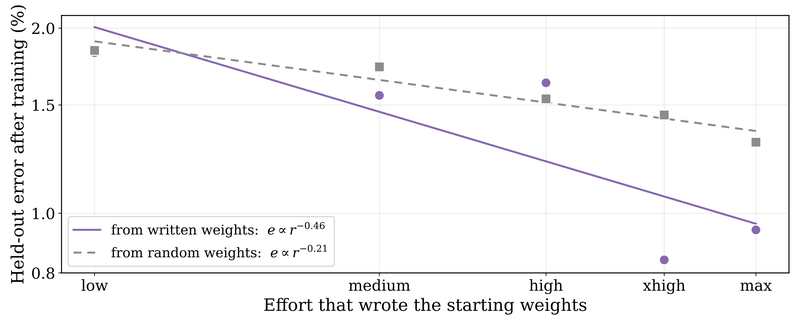


MNIST, 60,000 training and 10,000 test images. Best of five reasoning efforts; dashed lines are trained models.

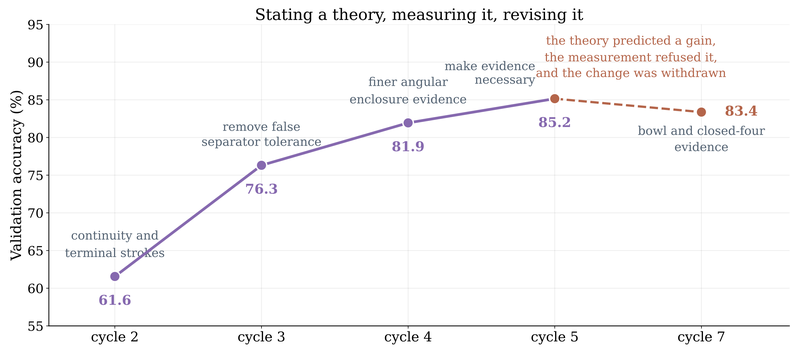
> Given no training images at all, a theory of how digits are written classifies most of them.

Reasoning as initialization

Networks initialized from written weights train to lower error, and the gap widens with reasoning effort.



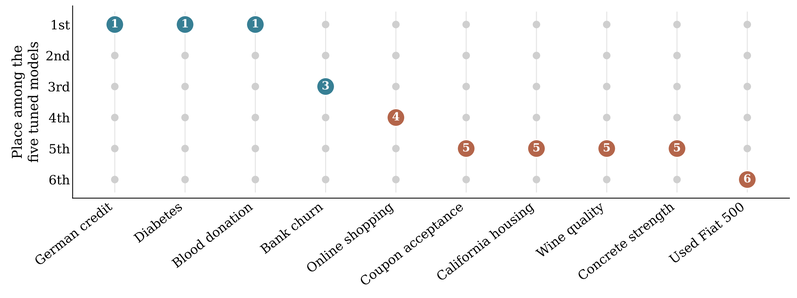
3 The loop the model runs



One digits session. Each label is the change the model argued for before measuring.

> It abduces a cause, deduces a change, and withdraws the change when the measurement refuses it.

4 The same protocol, ten fields

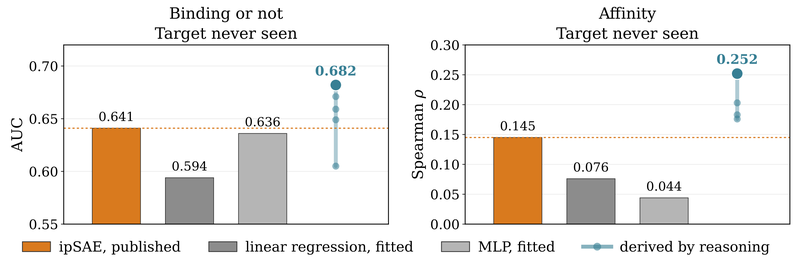


Ten tasks from TabArena, ranked against five tuned models on the same rows.

> Wins on small clinical tables, trails boosted trees on feature interactions.

5 Deriving a scientific score

Each session receives a structure predictor's error matrix and nothing else.



704 designed binders, holding out a whole target. Detection on the left, affinity on the right.

> A rule derived from physics beats the published score on a target it has never seen, and beats both fitted models.

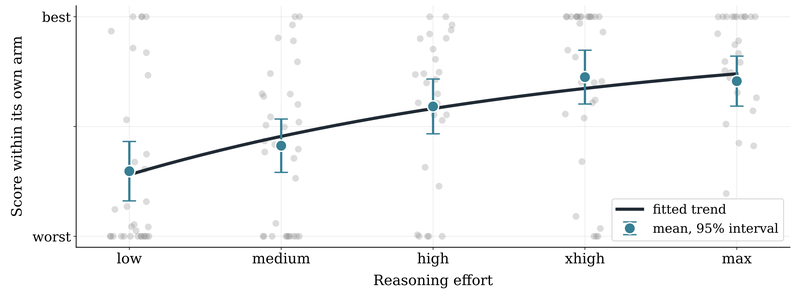
The theory it stated

Treat the bound complex as a harmonic well. Equipartition makes the interface stiffness the inverse square of its fluctuation, and a stiffer well is a deeper one. Dividing by the binder's own internal error removes the predictor, and the units with it. Association confines six rigid-body degrees of freedom, each worth half the thermal energy times the log of the stiffness.

$$\kappa = \left(\frac{\sigma_0}{\sigma_{\text{iface}}}\right)^2, \qquad \mathrm{p}K_{\mathrm{d}} = \mathrm{p}K_{\mathrm{d,ref}} + 3\log_{10}\kappa.$$

The three is six halved. Nothing is fitted.

6 Scaling reasoning effort



7 So how far can reasoning go?

- Deep learning dominates the domains that are data-rich and theory-poor.
- Those are exactly the domains whose first principles nobody has written down yet.
- As reasoning improves, we expect models to supply them, and stated theories to replace fitted predictors there.